

# Advancing Legal Reasoning in Algerian Law

RAG + Knowledge Graphs + Argument Mining → Citation-Faithful legal answers

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WHY  
01

## The Problem

WHY LEGAL QA IS HARD

In law, **fluent ≠ correct**. An answer is legally valid only if it is

sourced

article-cited

jurisdiction-bound

amendment-aware

— or refused

REAL QUESTIONS, REAL DEMANDS — AlgerianLegalBench

يَبْنِ الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي،  
وحدود سلطته عند غياب النص...

needs **exact articles** — Civil Code Art. 1–2

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل 2005؟

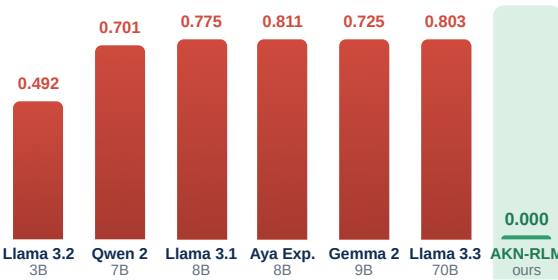
needs the law **as amended** — Ord. 05-02 → Art. 54

هل يحق لصاحب العمل في الجزائر فصل العامل بحرية تامة دون  
إبداء أسباب **(at-will)**؟

**foreign-doctrine trap** — Algerian law is silent → **abstain**

OFF-THE-SHELF LLMS FAIL THIS BAR

**Hallucinated-Citation Rate** SHARE OF CITED ARTICLES THAT DON'T EXIST



Up to **4 of 5** cited articles are **fabricated** — and the **70B** model is among the worst. **Scale doesn't fix law.**

**6 of 7** also import **foreign law** on >12 of 40 trap questions; Gemini 2.5 stays "clean" (0.008) only by **declining to cite at all**.

...AND ALGERIAN LAW IS A HARD CASE



### Arabic-first

232 / 244 benchmark Qs in Arabic



### Layered in time

1963–2025 · amendments rewrite articles



### Deep structure

code → chapter → section → article



### Low-resource

barely covered by legal-AI datasets

OBJECTIVES



### Structure the law

171 docs → AKN XML + legal KG



### Benchmark it

AlgerianLegalBench · 244 expert Qs



### Reason on evidence

typed retrieval + bounded recursion



### Verify or abstain

triple faithfulness gate

SIGNIFICANCE & IMPACT



### First citation-faithful QA for Algerian law

every claim resolves to a registered article — or the system abstains



### Reusable legal infrastructure

structured corpus + KG + benchmark for Algerian legal NLP



### A recipe for low-resource legal AI

evidence control beats parameter scale

HOW  
02

## The AKN-RLM Architecture

DOCUMENTS → STRUCTURE → VERIFIED ANSWER

Four zones, each repairing a failure mode of the last — a controlled evidence pipeline, not one model.

### 1 STRUCTURE THE LAW

171 files · 1963–2025  
FRBR article registry



Raw texts

PDF · TXT



Akoma Ntoso

XML · articles



Knowledge Graph

RDF · amend. chains

### 2 RETRIEVE EVIDENCE

RRF( BM25 + Dense )



Legal question

AR · FR



Classifier

routes 8 types

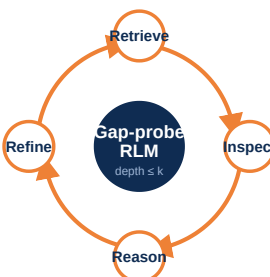


Hybrid retrieval

typed handlers

### 3 REASON RECURSIVELY

bounded · gap-driven  
depth-limited



- **Detects evidence gaps**

insufficient support → expand

- **Toulmin ADU mining**

on every cited article

- **One corrective retry**

then stops — no open recursion

✓ Evidence sufficient → verify & answer

### 4 VERIFY & ANSWER

triple faithfulness gate



Citation exists

vs registry



Jurisdiction OK

canary check



Claim supported

multiling. NLI



Verifiable answer

every claim → a registered article



Principled abstention

when the law is silent

PROOF  
03

## Results

FROZEN 244-QUESTION RUN

0.305

Citation  
F1

locked

deployable

✓ 0.000 HCR

· JIR

0.703

✓ Abstention

F1

1.64×

vs direct LLM

2.9×

vs determ. retr.

1.74×

vs minimal RAG



Citation F1 vs baselines

HIGHER = BETTER

Direct LLM

raw

0.056

Determ.

hybrid+rerank

0.105

Minimal RAG

dense top-5

0.175

Best LLM

Gemini 2.5

0.186

AKN-RLM

locked

0.305



Jurisdictional infection (JIR)

LOWER = SAFER

AKN-RLM

triple gate

0.000

Gemini 2.5

15 / 40 traps

0.375

Qwen 2

27 / 40 traps

0.675

Only AKN-RLM holds **0.000** hallucinated-citation & foreign-law rates. Reliability tracks **legal structure, not scale** — the largest baselines hallucinate most.



AlgerianLegalBench

244 QUESTIONS

23

CATEGORIES

8

QUERY TYPES

40

INFECTION TRAPS

232

ARABIC Q

12

FRENCH Q

.829

K AGREEMENT

Difficulty mix

Easy · Medium · Hard

23%

35%

42%



LIMITS

- Retrieval ceiling: R@10 = 0.258
- French under-served
- Abstention precision 0.654
- Metrics are proxies



NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review



Structured sources

+



Controlled retrieval

+



Recursive reasoning

+



Citation verification

=

THE TAKEAWAY

Trustworthy Legal AI

an **evidence-controlled system**, not an unconstrained generator.